

Time Series Analysis, Monte Carlo Simulation, and European Option Pricing of Apple Inc. Stock

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Abstract

This report analyzes Apple Inc. stock (AAPL) as the underlying asset for a computational finance application. The empirical analysis is based on daily historical data from 2016-01-01 to 2025-12-31, covering ten full calendar years. The project combines time series analysis, ARIMA modeling, Geometric Brownian Motion, Monte Carlo simulation, and European option pricing. First, the statistical properties of AAPL prices and log returns are examined using exploratory data analysis, descriptive statistics, stationarity tests, and an ARIMA benchmark model. Second, the estimated return and volatility parameters are used to simulate future AAPL price paths and to value European call and put options. The Monte Carlo option prices are compared with Black-Scholes prices to evaluate the accuracy of the simulation framework.

Contents

1	Introduction and Objectives	3
2	Data Acquisition	3
3	Data Cleaning and Preparation	4
4	Exploratory Data Analysis	4
4.1	AAPL Price Development	4
4.2	Daily Log Returns	5
4.3	Return Distribution	5
4.4	Rolling Mean	6
4.5	Rolling Volatility	6
4.6	Autocorrelation	7
5	Statistical and Time Series Characteristics	8
5.1	Descriptive Statistics	8
5.2	Stationarity	9
5.3	Trend and Volatility Clustering	9
5.4	Distribution Behavior	10
5.5	Implications for ARIMA and GBM	10
6	Time Series Modeling	11
7	Geometric Brownian Motion Framework	11
8	Monte Carlo Simulation	12
9	European Option Pricing	14
10	Conclusion and Limitations	16

1 Introduction and Objectives

Financial markets generate large amounts of time series data that can be analyzed using computational methods. In this project, Apple Inc. stock, traded under the ticker symbol AAPL, is selected as the underlying asset. Apple is one of the most actively traded large-cap technology companies and has a highly liquid stock and options market. This makes AAPL a suitable asset for a computational finance application involving time series analysis, Monte Carlo simulation, and European option pricing.

The empirical analysis is based on daily AAPL stock price data from 2016-01-01 to 2025-12-31. This period covers ten full calendar years and provides a sufficiently large number of observations for time series analysis. The selected daily frequency is appropriate for studying return behavior, volatility dynamics, stationarity, and potential time series models.

The objective of this first part of the project is to prepare and analyze the historical AAPL time series. More specifically, this part focuses on data acquisition, data cleaning, return calculation, exploratory data analysis, and the statistical characterization of the price and return series. The results of this analysis provide the empirical foundation for the later Geometric Brownian Motion framework, Monte Carlo simulation, and European option pricing.

The analysis is implemented in Python using a structured Jupyter Notebook. This is consistent with the computational finance approach discussed by Hilpisch (2019). The cleaned dataset and estimated return parameters are later used as inputs for the simulation and option pricing part of the project.

2 Data Acquisition

The historical data for Apple Inc. are retrieved from Yahoo Finance using the Python package `yfinance` (Aroussi, 2024; Yahoo Finance, 2026). The selected ticker symbol is AAPL. The dataset is downloaded at daily frequency and covers the period from 2016-01-01 to 2025-12-31. In the Python implementation, the end date is set to 2026-01-01 because the `end` argument in `yfinance` is treated as an exclusive boundary.

The raw dataset contains the standard daily market variables: open price, high price, low price, close price, adjusted close price, and trading volume. For the subsequent analysis, the adjusted closing price is selected as the main price variable. This is important because adjusted prices account for corporate actions such as dividends and stock splits. For a long-term stock analysis over ten years, using adjusted prices is more appropriate than using unadjusted closing prices.

The final raw dataset contains 2514 daily observations before return calculation. Since 2016-01-01 was not a trading day, the first available trading date in the dataset is 2016-

01-04. The final trading date is 2025-12-31. This is consistent with the intended ten-year sample period.

3 Data Cleaning and Preparation

After downloading the raw data, the dataset is inspected for column names, data types, missing values, and date order. The adjusted closing price is selected and renamed as the main price variable. The time index is converted to a proper date format and sorted in chronological order.

Missing values are removed from the selected price series. For daily financial data, missing calendar days usually correspond to weekends and market holidays and are therefore not part of the trading-day dataset. After cleaning and calculating returns, the final dataset contains 2513 observations.

Daily simple returns and daily log returns are calculated. The simple return is defined as:

$$R_t = \frac{S_t - S_{t-1}}{S_{t-1}},$$

where S_t denotes the adjusted closing price at time t . The daily log return is calculated as:

$$r_t = \ln \left(\frac{S_t}{S_{t-1}} \right).$$

Log returns are commonly used in financial time series analysis because they are additive over time and often have more stable statistical properties than price levels. The cleaned dataset contains the adjusted AAPL price, simple returns, and log returns. This dataset is saved as `aapl_clean.csv` and is used as the common dataset for the subsequent simulation and option pricing analysis.

4 Exploratory Data Analysis

4.1 AAPL Price Development

Figure 1 shows the adjusted AAPL price over the sample period. The price series displays a strong upward trend over the ten-year period. This visual pattern suggests that the raw price level is likely non-stationary, which is later examined formally using the Augmented Dickey-Fuller test.

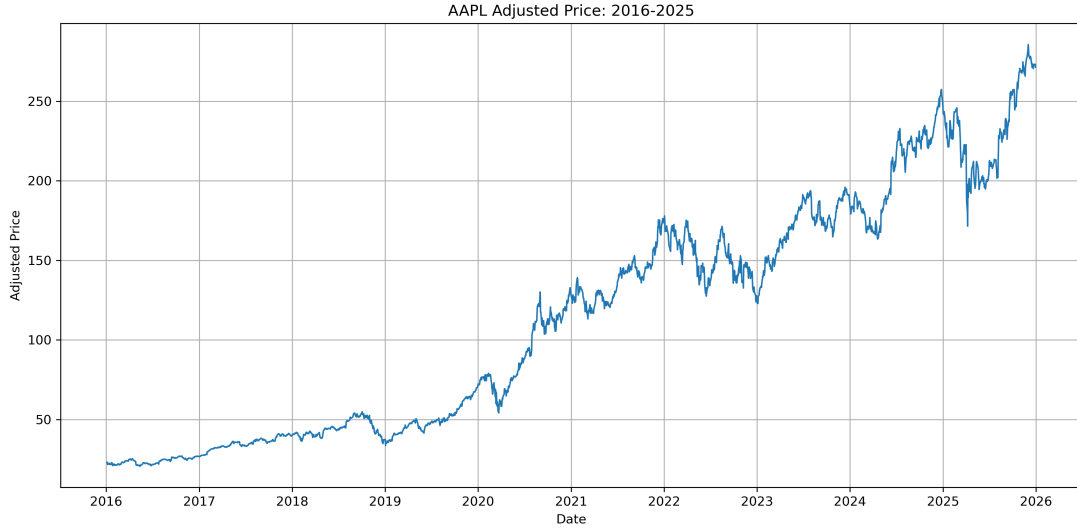


Figure 1: Adjusted AAPL price from 2016 to 2025

4.2 Daily Log Returns

Figure 2 shows the daily log returns of AAPL. In contrast to the price series, the return series fluctuates around zero. Several periods of increased return variation can be observed, which indicates that volatility is not constant over time.

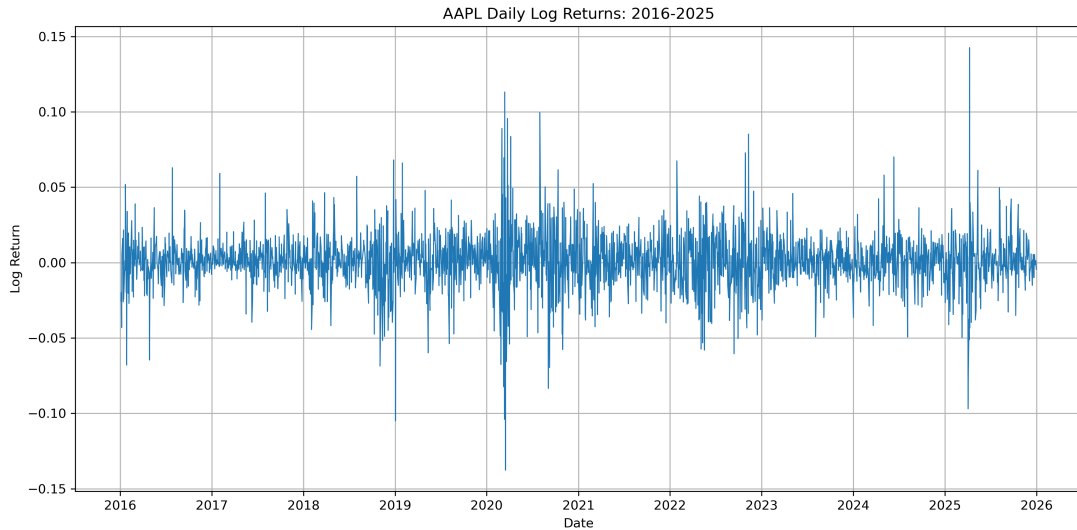


Figure 2: Daily AAPL log returns from 2016 to 2025

4.3 Return Distribution

Figure 3 shows the empirical distribution of daily AAPL log returns. Most observations are concentrated around zero, while extreme positive and negative returns are also present. This indicates that the return distribution may deviate from the normal distribution.

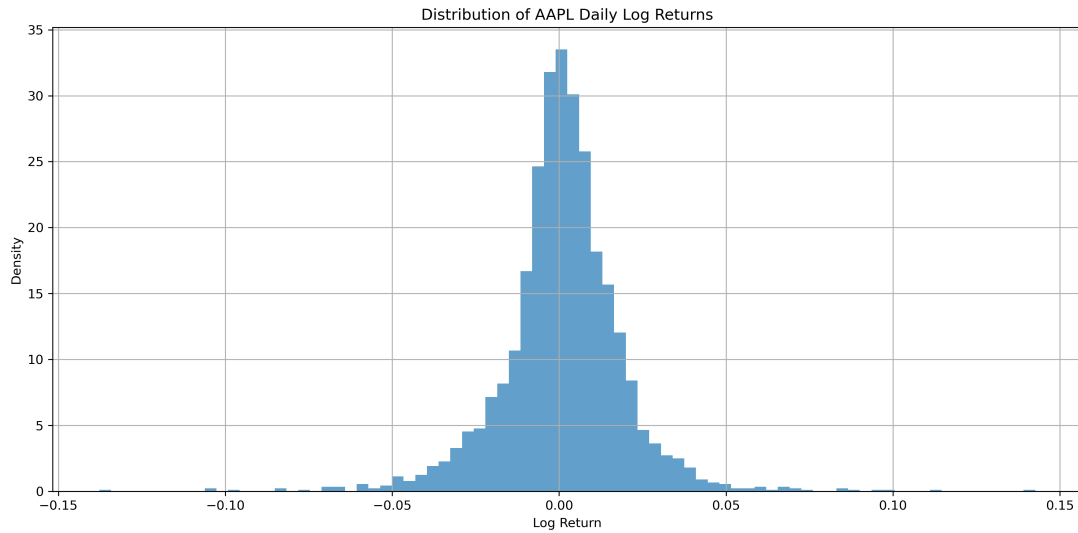


Figure 3: Distribution of daily AAPL log returns

4.4 Rolling Mean

Figure 4 presents the 30-day rolling mean of AAPL log returns. The rolling mean fluctuates around zero and remains small compared with the volatility of daily returns. This indicates that short-term average returns vary over time but do not dominate the overall return dynamics.

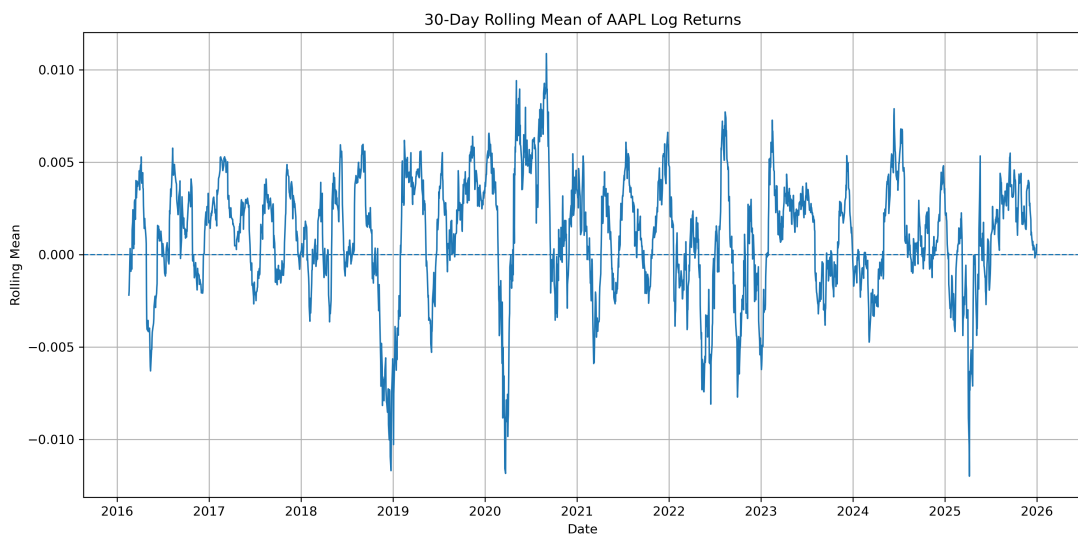


Figure 4: 30-day rolling mean of AAPL log returns

4.5 Rolling Volatility

Figure 5 shows the 30-day rolling volatility of AAPL log returns. The figure indicates that volatility changes over time and that periods of high volatility are followed by further

periods of high volatility. This is consistent with volatility clustering, which is a common feature of financial return series.

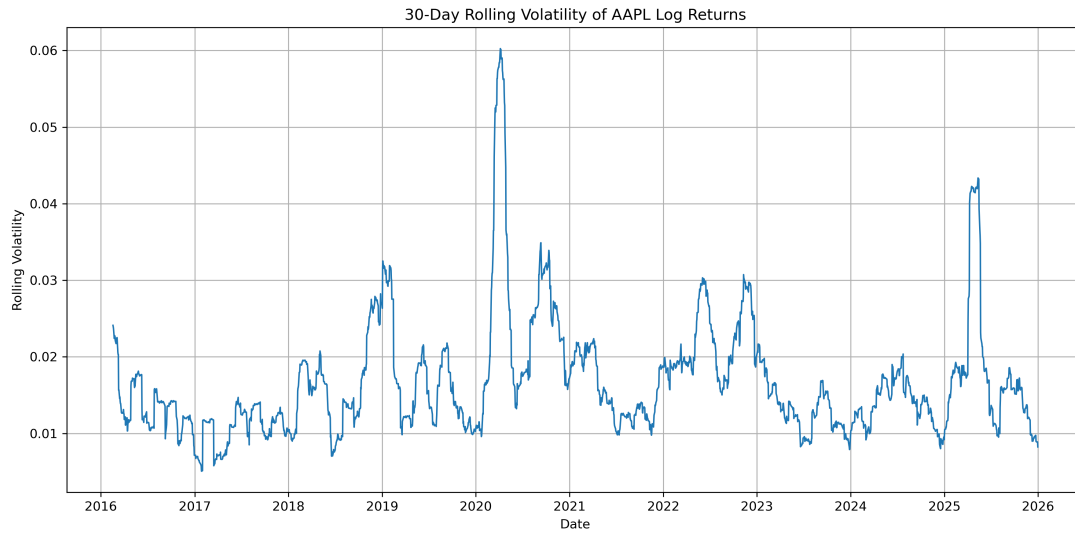


Figure 5: 30-day rolling volatility of AAPL log returns

4.6 Autocorrelation

Figure 6 shows the autocorrelation function of daily AAPL log returns. The ACF is used to examine whether current returns are linearly related to past returns. If most autocorrelation coefficients lie within the confidence bands, this suggests limited linear predictability in daily returns.

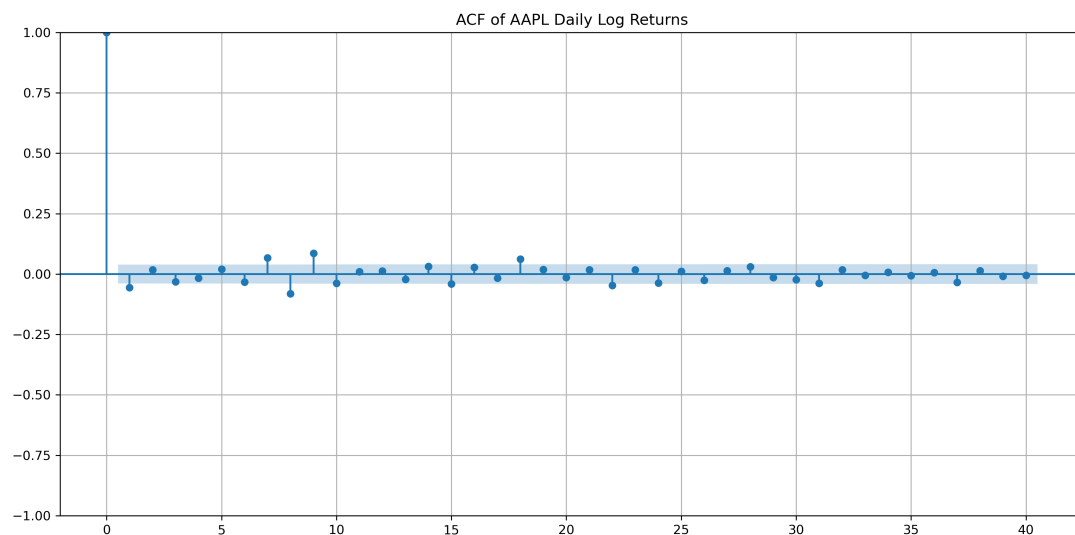


Figure 6: Autocorrelation function of AAPL daily log returns

5 Statistical and Time Series Characteristics

5.1 Descriptive Statistics

Table 1 summarizes the main descriptive statistics of AAPL daily log returns. The final cleaned dataset contains 2513 observations. The last observed adjusted AAPL price is 271.36. The daily mean log return is 0.000970, while the daily volatility is 0.018281. Annualizing these values with 252 trading days gives an annualized mean return of 0.244436 and an annualized volatility of 0.290207.

Table 1: Descriptive statistics of AAPL daily log returns

Metric	Value
Number of observations	2513
Start date	2016-01-05
End date	2025-12-31
Last AAPL price	271.35835
Daily mean log return	0.000970
Daily volatility	0.018281
Annualized mean return	0.244436
Annualized volatility	0.290207
Minimum daily log return	-0.137708
Maximum daily log return	0.142617
Skewness	-0.091838
Excess kurtosis	6.697244
Jarque-Bera statistic	4699.715555
Jarque-Bera p-value	0.000000

The positive average return indicates that AAPL generated positive average daily log returns over the sample period. However, the daily mean is small relative to daily volatility, which is typical for financial return series. The annualized volatility of approximately 29.02% indicates substantial risk and will be an important input for the later Monte Carlo simulation.

The skewness is slightly negative, suggesting a small degree of left-skewness in the return distribution. The excess kurtosis is strongly positive at 6.697244, indicating fat tails relative to the normal distribution. The Jarque-Bera p-value is effectively zero, so the null hypothesis of normally distributed returns is rejected. This result is consistent with the empirical observation that financial returns often display non-normality and extreme observations (Jarque and Bera, 1980).

5.2 Stationarity

The Augmented Dickey-Fuller test is used to evaluate whether the price and return series are stationary. The null hypothesis of the ADF test is that the series has a unit root and is therefore non-stationary (Dickey and Fuller, 1979).

Table 2: Augmented Dickey-Fuller test results

Series	ADF Statistic	p-value	Lags Used	Observations
AAPL Price	0.386286	0.980976	10	2502
AAPL Log Return	-16.221231	0.000000	8	2504

For the AAPL price series, the ADF p-value is 0.980976. Since this value is above the 5% significance level, the null hypothesis of a unit root cannot be rejected. Therefore, the AAPL price series is interpreted as non-stationary.

For the AAPL log return series, the ADF p-value is approximately zero. Since this value is below the 5% significance level, the null hypothesis of a unit root is rejected. Therefore, the AAPL log return series is interpreted as stationary. This result supports the use of log returns rather than raw price levels for statistical time series modeling.

5.3 Trend and Volatility Clustering

The price plot shows a clear upward trend over the sample period, which is consistent with the ADF result that the price level is non-stationary. In contrast, log returns fluctuate around a relatively stable mean close to zero.

The rolling volatility plot and squared return plot indicate that volatility is not constant over time. Periods of high volatility are visible, and large return movements tend to cluster together. This volatility clustering is a common empirical feature of financial time series and suggests that constant-volatility models are a simplification.

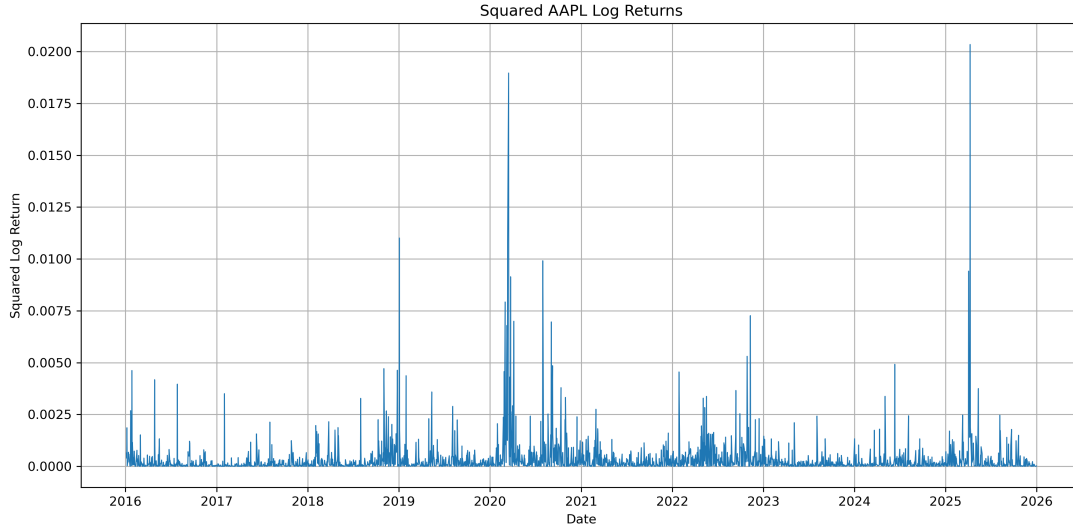


Figure 7: Squared AAPL log returns

5.4 Distribution Behavior

The histogram and descriptive statistics show that AAPL returns are not normally distributed. The strongly positive excess kurtosis indicates fat tails, meaning that extreme return observations occur more frequently than under a normal distribution. The Jarque-Bera test confirms this finding by rejecting normality.

This has important implications for the later simulation and option pricing analysis. Models based on normal returns or constant volatility may not fully capture the empirical characteristics of AAPL returns. Nevertheless, these models can still provide a useful benchmark if their limitations are clearly discussed.

5.5 Implications for ARIMA and GBM

The stationarity of the log return series suggests that time series models such as ARIMA may be applied to returns rather than price levels. The ARIMA framework is designed to model linear time dependence in stationary series and is commonly used as a benchmark in time series analysis (Box and Jenkins, 1970).

For the Monte Carlo simulation part, the Geometric Brownian Motion framework requires estimates of drift and volatility. The daily mean log return and daily volatility estimated in this section provide the empirical basis for calibrating the GBM model. However, the observed non-normality and volatility clustering indicate that the GBM assumption of constant volatility is a simplification of real market behavior.

6 Time Series Modeling

This section provides a benchmark time series model for the AAPL daily log return series. The stationarity analysis in the previous section shows that the AAPL price level is non-stationary, while the log return series is stationary. Therefore, the ARIMA model is estimated on daily log returns rather than on price levels.

The ARIMA model is commonly used for modeling stationary time series and follows the Box-Jenkins approach (Box and Jenkins, 1970). An ARIMA model is written as $ARIMA(p, d, q)$, where p is the autoregressive order, d is the differencing order, and q is the moving average order. Since the AAPL log return series is stationary, the differencing parameter is set to $d = 0$.

Several low-order ARIMA specifications are considered and compared using the Akaike Information Criterion (AIC). Among the tested models, $ARIMA(1,0,0)$ achieves the lowest AIC and is therefore selected as the benchmark model for the AAPL daily log return series. The selected model has an AIC of -12984.699 and a BIC of -12967.212.

Table 3: ARIMA benchmark model results

Metric	Value
Selected benchmark model	$ARIMA(1,0,0)$
AIC	-12984.699
BIC	-12967.212

The selected $ARIMA(1,0,0)$ benchmark model suggests that AAPL returns exhibit only limited short-term autoregressive dependence. In other words, current returns may be weakly related to recent past returns, but much of the variation in daily stock returns remains difficult to predict. This is consistent with the general empirical observation that daily financial returns are noisy and contain limited linear predictability.

The ARIMA model is therefore used as a statistical benchmark rather than as the main pricing model. Its purpose is to evaluate the linear time series structure of AAPL returns, while the later Monte Carlo simulation focuses on generating possible future stock price paths for option valuation.

7 Geometric Brownian Motion Framework

The Monte Carlo simulation in this project is based on the Geometric Brownian Motion framework. GBM is widely used in financial modeling because it describes stock prices as a stochastic process with continuous compounding, drift, volatility, and random shocks. It also forms the basis of the classical Black-Scholes option pricing framework (Black and Scholes, 1973; Merton, 1973).

The continuous-time GBM process is defined as:

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

where S_t is the stock price at time t , μ is the drift parameter, σ is the volatility parameter, and dW_t is the increment of a Brownian motion process.

For discrete-time simulation, the GBM process can be approximated as:

$$S_{t+\Delta t} = S_t \exp \left(\left(\mu - \frac{1}{2}\sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z_t \right),$$

where Z_t is a standard normally distributed random variable.

In this project, the historical annual drift is estimated as $\mu = 0.2444$, and the annualized volatility is estimated as $\sigma = 0.2902$. The drift captures the average historical return tendency of AAPL, while volatility measures the uncertainty of price movements. The estimated annualized volatility of approximately 29.02% indicates meaningful uncertainty and is a central driver of option value.

It is important to distinguish between historical simulation and risk-neutral option pricing. For simulating historical risk-return dynamics, the historical drift μ can be used. However, for European option pricing, the risk-neutral valuation framework requires the stock price process to grow at the risk-free rate rather than at the historical expected return. Therefore, the option pricing simulation uses the risk-free rate $r = 0.045$ as the drift.

Although GBM is a standard benchmark model, it relies on simplifying assumptions. In particular, it assumes constant volatility and normally distributed log returns. The empirical analysis in the previous sections shows that AAPL returns display fat tails and volatility clustering. Therefore, GBM should be interpreted as a useful benchmark rather than a perfect description of real market behavior.

8 Monte Carlo Simulation

Monte Carlo simulation is used to generate possible future price paths for AAPL based on the calibrated GBM model. Monte Carlo methods are widely used in financial engineering because they allow stochastic models and derivative payoffs to be evaluated numerically (Glasserman, 2004).

The simulation starts from the last observed adjusted AAPL price in the dataset. The initial stock price is $S_0 = 271.36$. The historical annual drift is estimated as $\mu = 0.2444$, and the annualized volatility is estimated as $\sigma = 0.2902$. The simulation horizon is one year, with 252 trading-day steps and 10,000 simulated price paths.

Table 4: Monte Carlo simulation input parameters

Parameter	Value
Initial stock price S_0	271.36
Historical drift μ	0.2444
Volatility σ	0.2902
Time horizon T	1.0
Number of time steps N	252
Number of simulations M	10,000

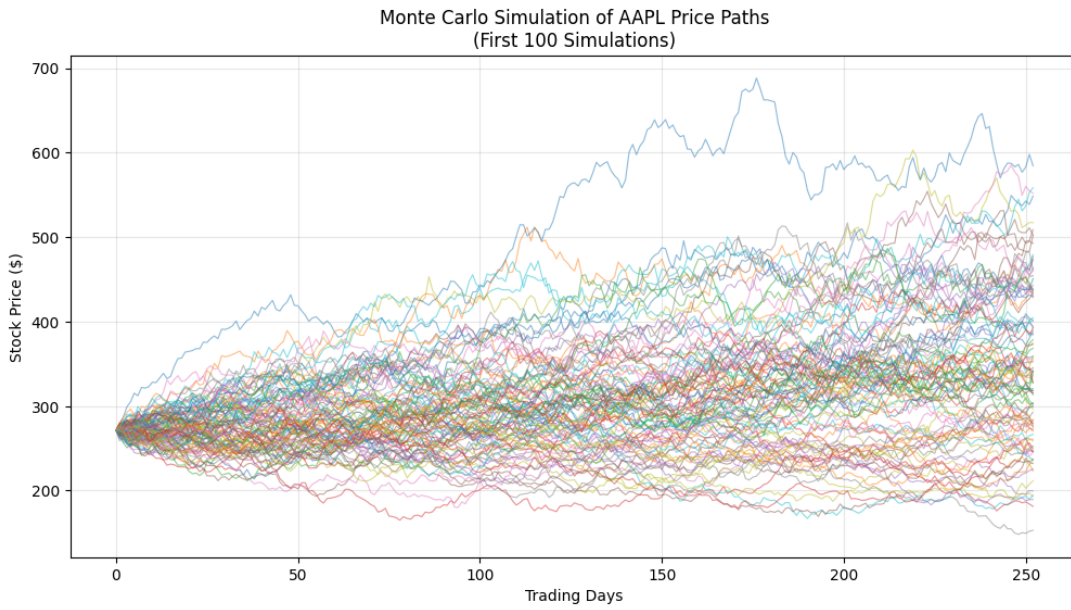


Figure 8: Monte Carlo simulation of AAPL price paths

Figure 8 shows the first 100 simulated AAPL price paths over the one-year horizon. All paths start from the same initial stock price, but they gradually diverge because random shocks accumulate over time. This illustrates the uncertainty captured by the Monte Carlo simulation and shows how the model generates a distribution of possible future stock price scenarios rather than a single deterministic forecast. The widening dispersion of the paths reflects the role of volatility in the GBM framework.

The simulated terminal stock price distribution is expected to be positively skewed under GBM. This means that most future prices are concentrated around moderate values, while there remains a non-negligible probability of substantially higher stock prices. This right-skewed distribution is a characteristic feature of lognormal stock price dynamics and is particularly relevant for call option valuation.

The Monte Carlo simulation therefore connects the historical time series analysis with the forward-looking option pricing framework. However, because GBM assumes constant

volatility, it cannot fully capture the volatility clustering observed in the historical AAPL return data.

9 European Option Pricing

The simulated AAPL price paths are used to estimate the values of European call and put options. A European option can only be exercised at maturity. The value of the option is determined by the expected discounted payoff under the risk-neutral valuation framework (Hull, 2018).

For a European call option, the payoff at maturity is:

$$\max(S_T - K, 0),$$

where S_T is the simulated stock price at maturity and K is the strike price. For a European put option, the payoff is:

$$\max(K - S_T, 0).$$

In the Monte Carlo approach, the payoff is calculated for each simulated terminal stock price. The option price is then obtained by taking the average payoff and discounting it back to the present:

$$C_0 = e^{-rT} \frac{1}{M} \sum_{i=1}^M \max(S_T^{(i)} - K, 0),$$

$$P_0 = e^{-rT} \frac{1}{M} \sum_{i=1}^M \max(K - S_T^{(i)}, 0),$$

where C_0 is the estimated call option value, P_0 is the estimated put option value, r is the risk-free rate, T is the time to maturity, and M is the number of simulations.

The option pricing inputs and results are summarized in Table 5. The strike price is set equal to the initial stock price, $K = S_0 = 271.36$, so the options are approximately at-the-money. The risk-free rate is set to $r = 0.045$, and the maturity is one year.

Table 5: European option pricing results

Parameter / Result	Value
Initial stock price S_0	271.36
Strike price K	271.36
Risk-free rate r	0.045
Time to maturity T	1.0
Number of simulations M	10,000
Monte Carlo call price	36.4829
Monte Carlo put price	25.1884
Black-Scholes call price	36.9526
Black-Scholes put price	25.0123
Call pricing difference	1.27%
Put pricing difference	0.70%

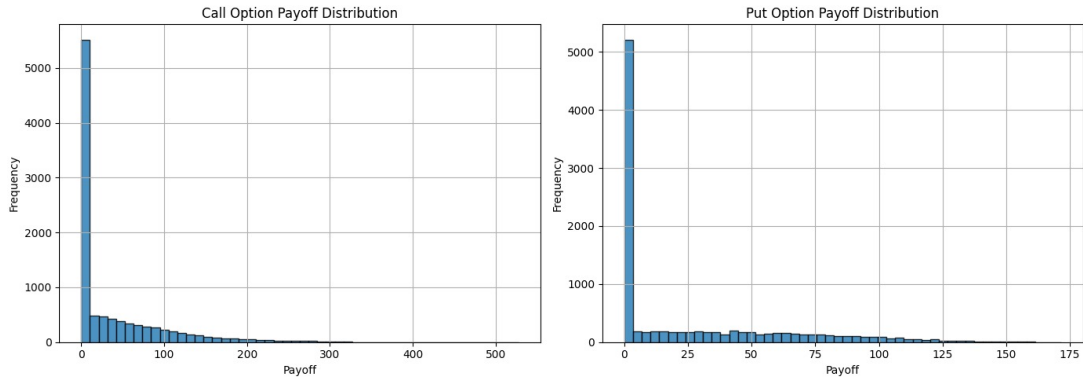


Figure 9: Distribution of simulated European call and put option payoffs

The Monte Carlo call price is 36.4829, while the corresponding Black-Scholes call price is 36.9526. The relative pricing difference is only 1.27%. For the put option, the Monte Carlo price is 25.1884 and the Black-Scholes price is 25.0123, giving a relative difference of 0.70%.

The close agreement between Monte Carlo and Black-Scholes prices provides evidence that the simulation framework was implemented correctly. Since both methods are based on the GBM assumption under risk-neutral valuation, the Monte Carlo estimates should converge toward the Black-Scholes prices when the number of simulations is sufficiently large. The small pricing differences indicate that 10,000 simulation paths are sufficient to produce stable option value estimates in this application.

Volatility is the primary driver of option value in this framework. Higher volatility increases the probability of extreme terminal stock prices, which raises the potential payoff of options. Therefore, the estimated annualized volatility of approximately 29.02% plays

a central role in both the Monte Carlo simulation and the option pricing results.

10 Conclusion and Limitations

This project combines empirical time series analysis, ARIMA modeling, Monte Carlo simulation, and European option pricing for Apple Inc. stock. The analysis is based on daily AAPL data from 2016-01-01 to 2025-12-31, covering ten full calendar years.

The empirical time series analysis shows that the AAPL price level is non-stationary, while the daily log return series is stationary. The return distribution displays fat tails and non-normal behavior, and the rolling volatility analysis indicates volatility clustering. These findings are consistent with common characteristics of financial return series.

The ARIMA analysis suggests that only a low-order model is needed to describe the limited linear dependence in daily AAPL returns. Based on the AIC comparison, the ARIMA(1,0,0) model is selected as the benchmark model, with an AIC of -12984.699 and a BIC of -12967.212. However, daily stock returns remain difficult to predict, so the ARIMA model is interpreted mainly as a statistical benchmark rather than as a strong forecasting model.

The estimated annual drift of AAPL is 0.2444, and the estimated annualized volatility is 0.2902. These parameters are used to calibrate the GBM model. Based on this model, Monte Carlo simulation generates possible future AAPL price paths and provides the basis for European option valuation.

For option pricing, the simulation is performed under the risk-neutral valuation framework using the risk-free rate of 0.045. The Monte Carlo option values are very close to the corresponding Black-Scholes prices. The Monte Carlo call price is 36.4829 compared with a Black-Scholes call price of 36.9526, while the Monte Carlo put price is 25.1884 compared with a Black-Scholes put price of 25.0123. These small pricing differences validate the Monte Carlo implementation.

The project demonstrates how historical data analysis can be connected with simulation-based derivative pricing. The approach provides a transparent computational workflow that links data preparation, statistical analysis, simulation, and valuation.

However, the model also has limitations. The GBM framework assumes constant volatility and normally distributed log returns. The empirical results show that AAPL returns exhibit fat tails and volatility clustering, meaning that the GBM assumptions are only approximations. In addition, the simulation results depend on historical parameter estimates and selected option pricing inputs, such as strike price, maturity, and risk-free rate.

Future improvements could include using GARCH-type models, stochastic volatility models, or jump-diffusion models to better capture volatility clustering and extreme market movements. The option pricing framework could also be extended by analyzing different strike prices, maturities, and market-implied volatilities.

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